UTM
UNIVERSITI TEKNOLOGI MALAYSIA

# Application Development
# SCSJ 3104

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# Application Development

- Application Development is a comprehensive service learning course which requires student to solve a real community problem by developing an application. Students will learn how to practice design thinking, adopting Agile development methodology. This involves an iterative process starting from community engagement, requirement elicitation and analysis, design solution, application construction and iterative verification process. Students are required to do reflection on the outcome of the project. In this course students should be able to develop their soft skills such as leadership, team collaboration, documentation process and communication skill.

# Learning Outcomes

- Identify community problem, analyse requirement, and propose solution in a well-structured proposal.
- Work in a team to develop an application based on the standard of software application development process.
- Demonstrate communication skill on community engagement and presentation

# Delivery Method




Online-based
Offline-based
Task-based

Proposal
Project planning
Sprint 1-4
Prototype
Final Report
Presentation

Webex Meetings

GitLab

Trello

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# Teaching PLANNING

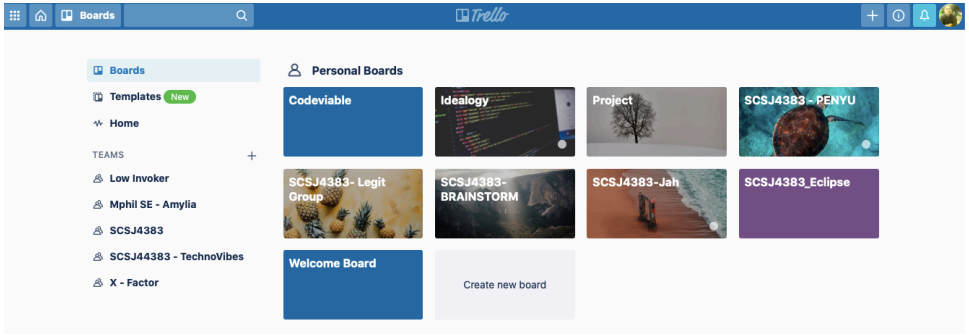
| Activity | PIC | Lesson Plan | Tools | Delivery |
|---|---|---|---|---|
| Online Teaching | Lecturer | Teaching via Webex application<br><br>Session 1 – 40 minutes online teaching | 1. Webex application<br>2. Trello | 1. Online<br>2. Each week will have 2 lesson planning |
| | Students | Session 2 – 1-hour Sprint review/stand-up meeting (group-based) | | |
| | Lecturer and Students | Session 3 - 20 minutes online discussion on sprint/project | | |
| Offline Teaching | Lecturer | 15-30 minutes teaching video | UTM E-Learning | Upload to UTM E-Learning |
| Task-Based | Students | Development | 1. Trello<br>2. Gitlab | Trello (Collaborative learning – lecturer will participate closely in the group discussion) |

# Assessment PLANNING

| Assessment | Approaches | Tools | Delivery |
|---|---|---|---|
| Proposal | Collaborative Learning<br>Marks will be given based on:<br>1. Content<br>2. Activeness of student discussion via tools<br>3. Peer assessment | 1. Trello (Project management)<br><br>2. Gitlab (Code versioning) | Group-Based<br><br>1. Trello<br>2. Gitlab<br>3. UTM E-Learning |
| Software Project Plan | | | |
| Sprint (1-4) | | | |
| Prototype | | | |
| Final Report | Online submission | UTM E-Learning | Group |
| Final Presentation | 1. Online<br>2. Peer assessment | Webex | Group |

# Assessment

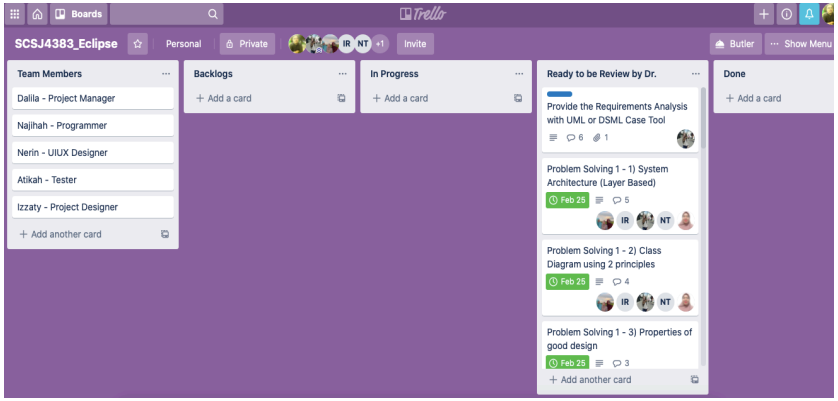
## Trello tools



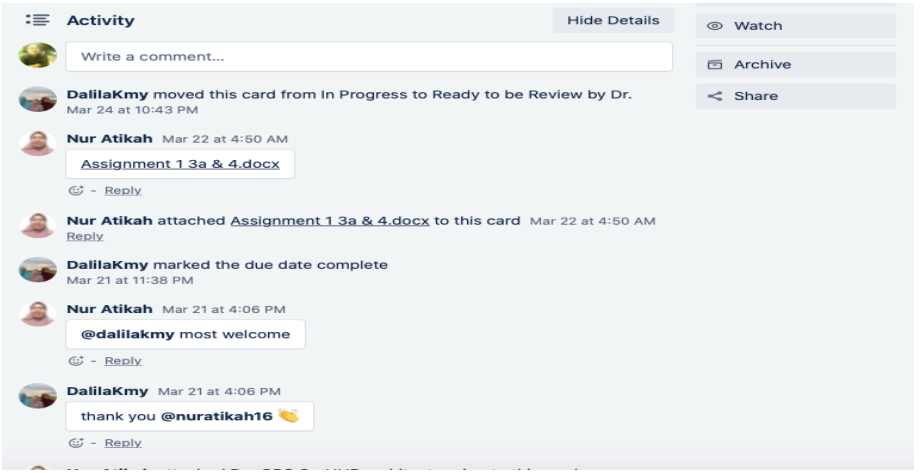

## Example – Task for assignee



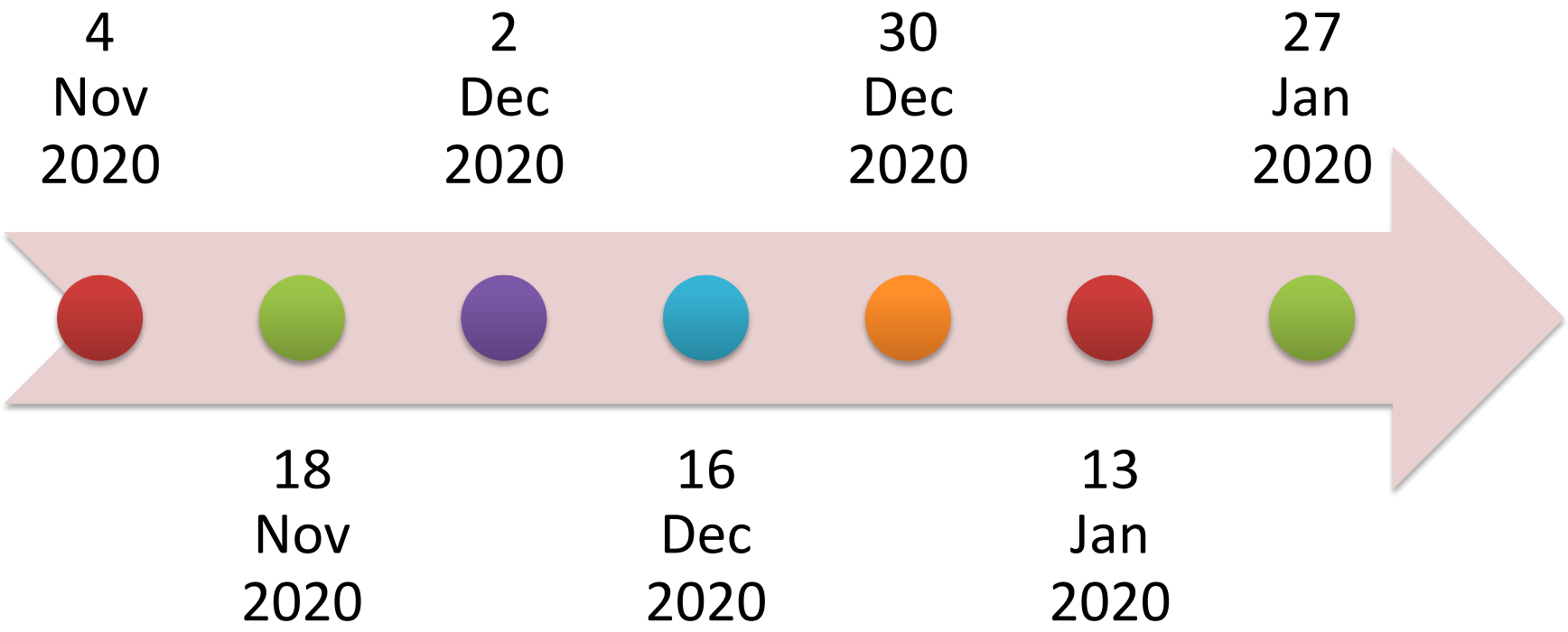

## Example - group




## Example – discussion



# Important Date.....



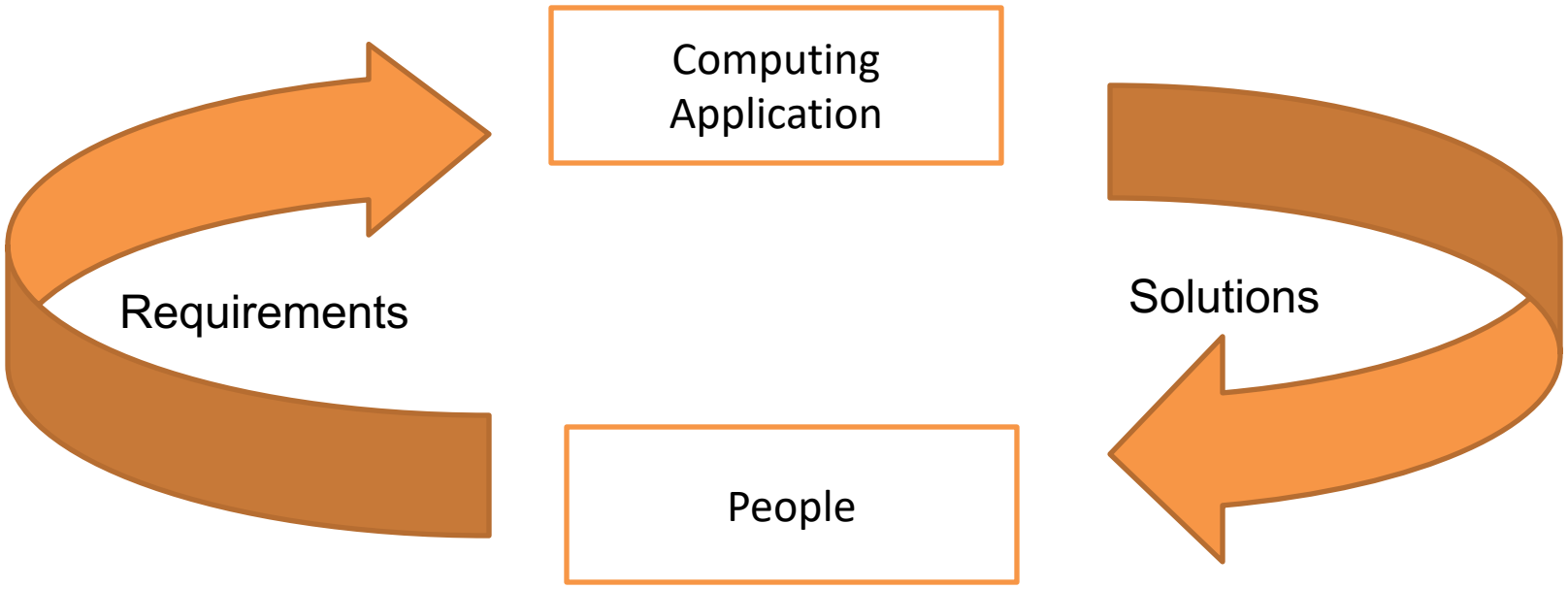
Individual – 20%

Group – 80%

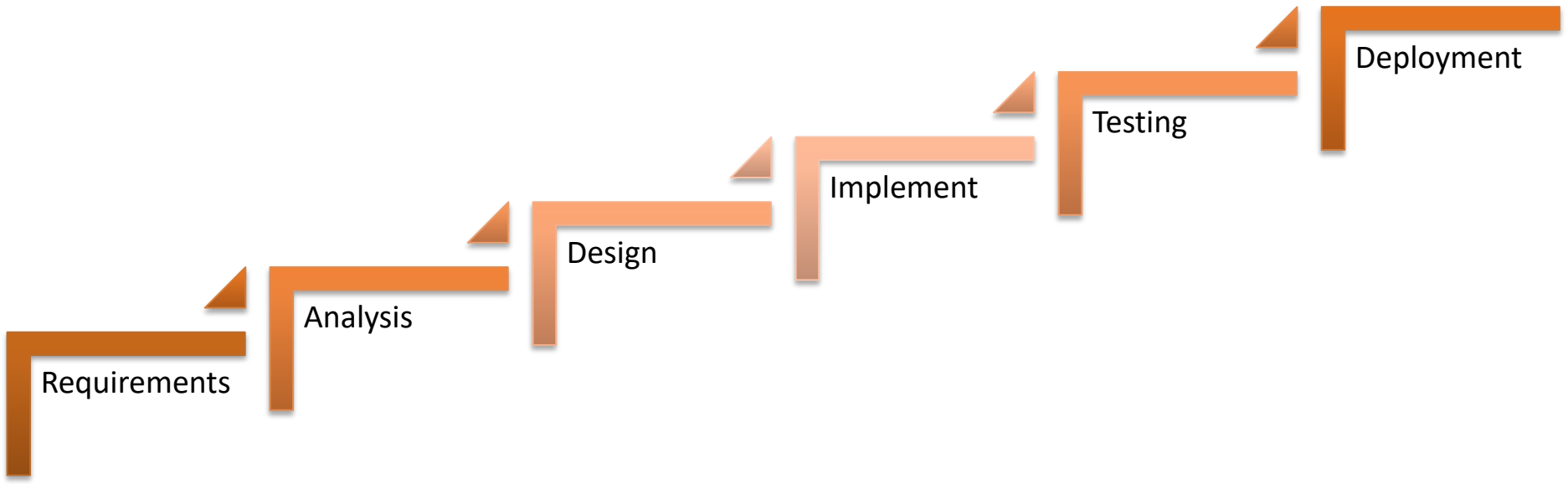
# Implementing Design Thinking Approach in the Application Development

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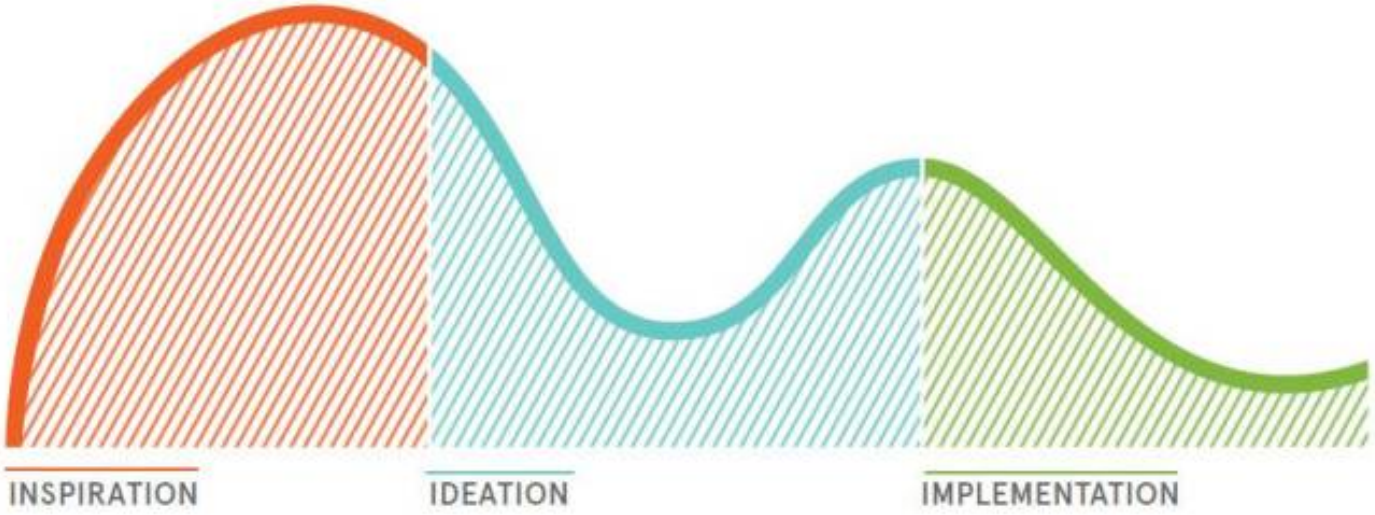
# What is the application ?



# Development Process



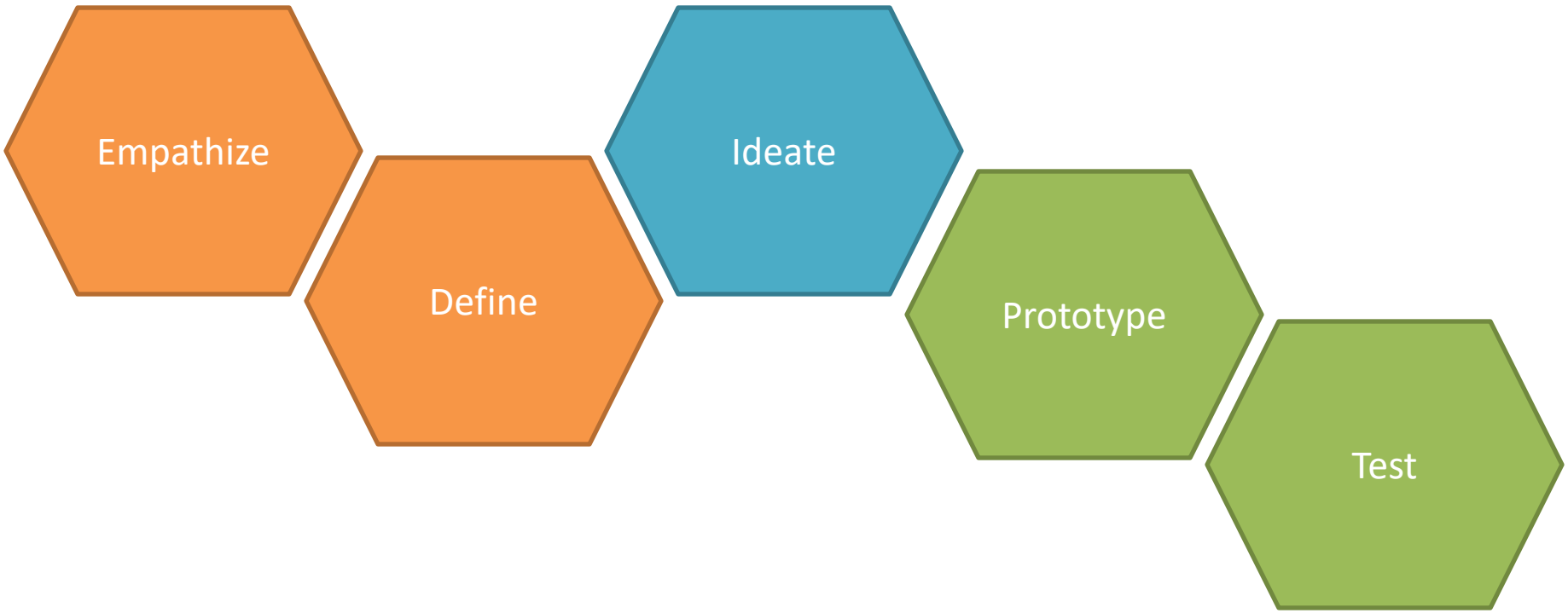
# HUMAN-CENTRED DESIGN



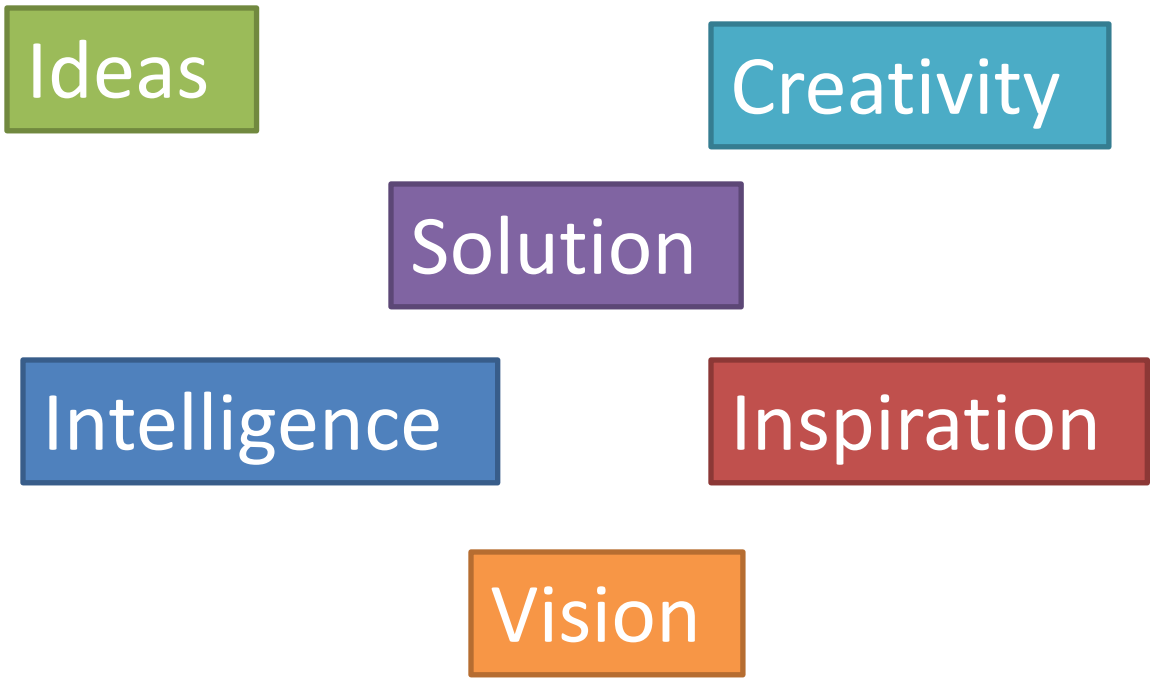
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# Design Thinking

# The Process



# INNOVATION


Ideas
Creativity
Solution
Intelligence
Inspiration
Vision